Deep Learning Course

Final Project

COLORECTAL CANCER TISSUE CLASSIFICATION

From Dense Networks to Vision Transformers

Medical Image Classification with PathMNIST

Progress check:	March 12, 2026
Deadline:	March 22, 2026 before 11:00 PM
Submission:	Jupyter Notebook (.ipynb) with all outputs
Defense:	Oral examination on submitted code

Academic Year 2025–2026

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Context

You will build and compare deep learning models to classify colorectal cancer histology images into 9 tissue types, applying everything learned in Labs 1–3. The project progressively increases in complexity: from a basic dense network (Lab 1), through convolutional architectures with regularization (Lab 2), to Transformers and transfer learning (Lab 3).

Dataset: PathMNIST from the MedMNIST benchmark — 107,180 colorectal cancer histopathology patches (28×28, RGB), sourced from the NCT-CRC-HE-100K dataset.

Classes:

Label	Tissue Type	Label	Tissue Type
0	Adipose	5	Smooth Muscle
1	Background	6	Normal Mucosa
2	Debris	7	Cancer Stroma
3	Lymphocytes	8	Tumor Epithelium
4	Mucus		

Clinical relevance: Automated tissue classification from histology slides can assist pathologists in cancer diagnosis, help quantify tumor composition, and accelerate analysis of whole-slide images.

Reminder

All models must be implemented in **PyTorch** (you may use any PyTorch-based library). All plots must have titles, axis labels, and legends. Every part must include a **written analysis** in your own words. **You will be asked to explain your code during an oral defense.**

1 Data Exploration

Load the PathMNIST dataset and understand its structure before building any models.

Perform data visualization and descriptive statistics on the dataset. Show that you understand what you are working with before building any model.

Question

Q1.1 — Look at several images from the *Debris* class and several from *Background*. Describe in your own words what visual differences you observe between them. Include at least one specific observation about color or texture.

Question

Q1.2 — Pick any single image from the training set and report its **pixel intensity statistics**: mean and standard deviation per channel (R, G, B). Are these values close to ImageNet statistics? Show the computation.

2 Dense Network Baseline

Build a multi-layer perceptron to establish a baseline.

- At least 2 hidden layers with dropout.
- Reach $\geq$ **55%** test accuracy.
- Plot training curves (loss and accuracy for train and validation).
- Show the confusion matrix on the test set.

Question

Q2.1 — Report your model's **exact test accuracy** and the **exact validation loss at epochs 1, 10, and at your final epoch**. Do not round — copy the values from your logs.

Question

Q2.2 — Look at the confusion matrix. Identify the single cell (predicted class, true class) with the **highest number of misclassifications**. Report the exact count and explain why this confusion might occur by looking at example images from both classes.

3 CNN from Scratch

Design and train a convolutional neural network with proper regularization.

- At least 3 convolutional blocks with batch normalization and dropout.

- Data augmentation with a justified choice of transformations.
- Reach $\geq$ **75%** test accuracy.
- Compare results **with vs. without** augmentation (same model, same hyperparameters, two runs).

Question

Q3.1 — Run your CNN **without** any augmentation for 40 epochs. At which epoch does the gap between training accuracy and validation accuracy exceed 15 percentage points for the first time? Report the exact epoch number and both accuracy values.

Question

Q3.2 — Histology images have a property that natural photographs do not: they have no canonical orientation (a tissue patch looks equally valid upside-down or mirrored). Name one common augmentation that exploits this property and one augmentation from natural image classification that would be **harmful or meaningless** for histology. Explain both choices.

Question

Q3.3 — Print the **total number of trainable parameters** in your CNN. Then, manually compute the number of parameters in your first convolutional layer (show the formula and calculation). Verify that it matches the model summary.

4 Transfer Learning with ResNet-18

Fine-tune a pretrained ResNet-18 on PathMNIST.

- Use ImageNet-pretrained weights.
- Handle the input size mismatch appropriately.
- Reach $\geq$ **85%** test accuracy.
- Run **two experiments**: (a) fine-tune only the classification head with all other layers frozen, (b) fine-tune the entire network. Report both results.

Question

Q4.1 — Report the **exact test accuracy** for both experiments (frozen vs. full fine-tuning). Which performed better, and by how many percentage points?

Question

Q4.2 — You are resizing 28×28 images to 224×224. This is an 8× upscale. Describe one concrete negative consequence of this extreme upscaling on what the model "sees." Suggest an alternative strategy that might reduce this problem.

Question

Q4.3 — A classmate claims that transfer learning works here because "ImageNet contains medical images." Is this claim correct? Give the actual reason transfer learning helps and be specific about what kind of features the early layers of a pretrained ResNet have learned.

5 Vision Transformer

Implement a Vision Transformer (ViT) **from scratch**. No pretrained ViT is allowed.

- Patch embedding with a learnable CLS token and positional embeddings.
- Use `nn.TransformerEncoder` for the self-attention layers.
- Classify using the CLS token output.
- Report and justify all hyperparameter choices.

Question

Q5.1 — Your images are 28×28. If you choose patch size 7, how many patches do you get? What if you choose patch size 14? For the patch size 14 case, explain concretely why the Transformer's self-attention mechanism would have very little to work with, and predict whether accuracy would be higher or lower. Then **run both** and report the actual results.

Question

Q5.2 — Remove the positional embeddings from your ViT entirely (set them to zero or remove them) and retrain. Report the accuracy drop. Explain why this drop is expected by connecting it to what positional embeddings encode and why spatial information matters for tissue classification.

Question

Q5.3 — Compare the number of parameters in your ViT to your CNN from Part 3. Which has more, and by what factor? Given this, explain why the ViT might still underperform the CNN on this dataset.

Deliverables: complete ViT implementation, training curves, test accuracy, written analysis.

6 Grad-CAM Interpretability

Implement Grad-CAM and apply it to your CNN (Part 3) or ResNet (Part 4) to visualize which image regions drive the model's predictions.

- Implement Grad-CAM yourself using PyTorch hooks. No external Grad-CAM library is allowed.
- The algorithm: forward pass → capture last conv layer activations → backward pass → capture gradients → weight activations by mean gradient per channel → ReLU → normalize.
- Visualize heatmap overlays for **at least 3 different tissue types**.
- Include at least 1 example where the model predicts **incorrectly**.

Question

Q6.1 — For a correctly classified *Tumor Epithelium* image and a correctly classified *Normal Mucosa* image, describe in concrete visual terms what is different about the two Grad-CAM heatmaps. Where on the image is the activation concentrated in each case? Refer to specific spatial patterns (e.g., "the heatmap highlights the center / edges / clustered regions / scattered spots").

Question

Q6.2 — Pick one of your misclassified images. Apply Grad-CAM for **both** the predicted class and the true class. Show both heatmaps side by side. What does the comparison tell you about why the model was confused?

7 Final Comparison and Analysis

7.1 Summary Table

Create a table comparing all 4 models: architecture, parameter count, test accuracy, and approximate training time.

7.2 Per-Class Analysis

For your best model, identify the top 3 most confused class pairs using the confusion matrix and per-class F1 scores. Explain why these confusions make sense visually or medically.

7.3 Written Analysis

Answer each of the following in **at least 5 sentences**:

1. **Architecture trade-offs:** Compare MLPs, CNNs, transfer learning, and ViTs for medical image classification. When would you choose each approach?

2. **Data efficiency:** Which models were most and least data-efficient? How did image resolution matter?
3. **Clinical deployment:** If deploying one model to assist pathologists, which would you pick and why? What additional steps would be needed before real clinical use?
4. **Ethics:** What are the risks of AI-based cancer tissue classification? Consider false negatives, class imbalance, dataset bias, and interpretability.

Question

Q7.1 — Plot all 4 models' validation accuracy curves on a **single graph**. At which epoch does each model first surpass 50% validation accuracy? Report the exact epoch for each.

Question

Q7.2 — Describe one unexpected result or behavior you encountered during this project that surprised you. What did you expect to happen, what actually happened, and how did you investigate or resolve it?

Question

Q7.3 — A colleague suggests using **random label smoothing** (replacing one-hot labels with uniform random vectors) to improve generalization. Without running the experiment, explain whether this would help or hurt, and why. Be specific.